

Kevin Tan

Diamond Bar, CA · k.tann.578@gmail.com · (909) 610-5135

[linkedin.com/in/kevinwtan](https://www.linkedin.com/in/kevinwtan) · github.com/ChocoKnight · chocoknight.github.io

Technical Skills

Languages	Python, TypeScript, JavaScript, Java, C#, SQL, Golang
Technologies	Git, Docker, Unix, MySQL, PostgreSQL, React, Redis, Azure
AI / ML	Large Language Models, Retrieval Augmented Generation (RAG), Neural Networks, Transformer Models, Machine Learning, Deep Learning, PyTorch, scikit-learn

Work Experience

Tax Relief Advocates

October 2025 - Present

Software Engineer | Python, TypeScript, SQL

Irvine, CA

- Developed an automated IRS eService pipeline using a high-performance asynchronous architecture and **Redis** background workers to enable parallel Secure Object Repository (SOR) mailbox scraping, replacing manual retrieval and tracking processes
- Architected a decoupled storage system for IRS transcripts, leveraging **Azure Blob Storage** for high-availability file hosting and **PostgreSQL** for high-speed metadata indexing, optimizing file lookup across thousands of records
- Engineered a high-performance IRS debt engine utilizing dictionary-based pre-indexing to calculate Collection Statute Expiration Dates (CSED) with **O(n)** efficiency, modeling complex multi-year tolling scenarios to project various taxpayer saving plans
- Architected a modular relational schema for the 433F computation engine, refactoring monolithic tables into normalized **PostgreSQL** structures to provide a scalable foundation for dynamic data entry and complex multi-page form generation
- Implemented **AES-256 envelope encryption** for sensitive PII and FTI (Federal Tax Information), utilizing Azure Key Vault for secure key management and automated rotation to ensure enterprise-grade data security
- Conducted API stress testing to identify a 100 requests/second throttling threshold, optimizing request consistency and asynchronous handling to maximize data ingestion rates under strict IRS rate limits
- Co-authored the **engineering guidelines** and spearheaded a modular pull request strategy to improve code maintainability, conducting rigorous technical reviews across microservices to ensure production stability

Infios

June 2025 - August 2025

Software Engineering Intern | Python | Large Language Models

Remote

- Engineered a **Retrieval-Augmented Generation (RAG) system** using **OpenAI GPT models** and **PostgreSQL** to automate internal technical support, significantly reducing manual documentation lookup time and decreasing repetitive support requests between teams
- Developed a custom hierarchical Markdown parser that preserved document structure through header-level chunking and parent-child relationships, increasing the contextual relevance of retrieved data
- Implemented end-to-end observability with **Langfuse** to monitor similarity scores, manage chat history context, and enforce internal safety guardrails for all LLM interactions

Education

California Polytechnic State University, San Luis Obispo

September 2021 - June 2025

B.S. Computer Science - Data Engineering Concentration

GPA: 3.732

Related Experience

Distributed Machine Learning Image Recognition System

May 2025 - June 2025

Distributed Computing, Deep Learning | Python, Golang

San Luis Obispo, CA

- Developed a leader-worker distributed engine in **Golang** to parallelize image classification, training a **Vision Transformer** optimized with class-weighted loss and data augmentation to reach 0.99 recall on highly imbalanced datasets